

Project Report: AI-Driven Parking Intelligence for Congestion Mitigation

1. Problem Analysis: The Invisible Gridlock

Operational Challenges On-street illegal parking and spillover parking—concentrated near commercial hubs, metro stations, and event venues—act as the primary catalysts for urban gridlock. From an architectural perspective, the mechanism of this congestion is a drastic reduction in effective carriageway capacity. A single vehicle illegally parked on a three-lane road does not simply reduce capacity by 33%; it creates a "bottleneck effect" where merging maneuvers and lane friction disrupt laminar flow, causing "spillover" gridlock that propagates to adjacent intersections and halts the broader network.**Current Enforcement Constraints** The Bangalore Traffic Police (BTP) currently operates under significant technical debt regarding parking management:

- **Reactive Patrol-Based Approach:** Enforcement relies on manual observation, which is inherently intermittent and lacks the scalability to cover the urban expanse.
 - **Absence of Impact Visualization:** There is no existing heatmap correlating specific parking violations with real-time congestion data, leaving the relationship between parking and flow unquantified.
 - **Prioritization Deficit:** Authorities lack the ability to rank violations by their "congestion weight." Currently, a vehicle blocking a high-throughput arterial road is often treated with the same urgency as one on a low-traffic side street, leading to inefficient resource allocation.
- The AI Objective** The core mission is to transition from manual, reactive patrolling to a targeted, data-driven enforcement framework. By utilizing computer vision for hotspot detection and traffic flow quantification, the solution aims to identify high-impact violation zones and prioritize them for immediate intervention.

2. Data Strategy & Constraints

Strict Dataset Usage Protocol **Warning: Adherence to HackerEarth protocols is mandatory. Participants must use ONLY the provided datasets for Problem Statement 1. The use of external datasets for training or ground truth is strictly prohibited and will result in immediate disqualification.****Data Inventory** The following data assets are earmarked for specific roles within the architecture:| Dataset Component | Role in Project | Specific Application || ----- | ----- | ----- || **Problem Statement 1 Dataset** | Ground Truth | Primary source for training detection models and quantifying hotspots. || **Surveillance Image Data** | Input Stream | Real-world image data used for vehicle localization and violation identification. || **Validation Data** | Performance Benchmarking | Used for calculating mAP and F1-scores to ensure model reliability. |

Data Integrity Measures To ensure high detection accuracy under diverse urban conditions, the preprocessing pipeline must incorporate:

- **Image Quality Normalization:** Global scaling of pixel intensities to standardize inputs.
- **Environmental Robustness:** Implementation of **Histogram Equalization** and **De-hazing algorithms** to mitigate the effects of low light, heavy rain, and motion blur as specified in the "Tasks" section.

- **Shadow Suppression:** Filtering techniques to prevent false-positive vehicle localization caused by high-contrast shadows.

3. Technical Architecture for AI-Driven Hotspot Detection

Detection Pipeline The Computer Vision workflow is structured into three discrete stages:

- **Vehicle and Road User Localization:**
- Multi-class detection of actors including vehicles (cars, LCVs, HCVs), riders, drivers, and pedestrians.
- Bounding box generation with high-confidence classification to distinguish between different modes of transport.
- **Violation Identification:**
- **Temporal Threshold Analysis:** To distinguish "Illegal Parking" from "stopped traffic," the system applies a velocity-over-time heuristic. If a vehicle remains stationary in a restricted zone for a duration exceeding the defined threshold, it is flagged as a violation.
- Logic filters specifically for illegal parking, ignoring transient stops like red-light dwell times.
- **License Plate Recognition (LPR):**
- Integration of OCR (Optical Character Recognition) to extract alphanumeric registration details from localized plates, creating a searchable record for enforcement.

Hotspot Generation Logic The transition from individual detection points to geographic intelligence is achieved through **Kernel Density Estimation (KDE)**. By mapping coordinate-tagged violation events, the system converts discrete points into continuous density surfaces (polygons). These "Violation Heatmaps" allow the system to identify persistent clusters where unauthorized parking is most frequent, rather than reacting to isolated incidents.

4. Traffic Flow Impact Quantification

Quantification Methodology We define the **Congestion Impact Score (CIS)** to measure the severity of a violation. The CIS is calculated as the product of the **Violation Duration** and the **Road Class/Throughput Volume**. By correlating the KDE-generated hotspots with road capacity data, the AI quantifies exactly how much throughput is lost due to specific parking behaviors, allowing BTP to prioritize the most disruptive hotspots. **Performance Metrics** The system's efficacy will be strictly evaluated using the following metrics:

- **Accuracy:** Overall detection correctness.
 - **Precision and Recall:** Minimizing false positives while ensuring no high-impact violations are missed.
 - **F1-score:** The balance between precision and recall.
 - **mAP (mean Average Precision):** The primary metric for multi-object localization performance.
- Analytical Reporting (Evidence Generation)** To facilitate legal enforcement, the system generates comprehensive evidence packages:
- **Annotated Evidence:** Images with highlighted violations, bounding boxes, and extracted LPR data.
 - **Metadata Logging:** Every event is stored with a high-precision timestamp and geographic metadata.

- **Trend Reporting:** Searchable records and summary reports for post-event learning.

5. Prototype Submission & Implementation Guide

Submission Checklist The submission window closes on **June 21, 2026, at 11:59 PM IST** .

Ensure the following tasks are completed:

- Access the HackerEarth platform via laptop/desktop (mobile participation is not supported).
- Click the **"Submit Now"** button and populate all mandatory fields.
- Validate that all uploaded files (code, concept notes, models) are under the **50 MB limit** per file.
- Execute the **"Publish"** action to finalize your entry.
- **Roadmap to the Onsite Finale**
- **Shortlisting:** A joint jury from the **Bangalore Traffic Police (BTP)** and **Flipkart** will evaluate the prototypes to select the top 10 teams.
- **Grand Finale:** The shortlisted teams will be invited to the onsite finale on **July 3, 2026** , at **Flipkart HQ, Bengaluru** .
- **Prize Eligibility:** To remain eligible for the **₹500,000 prize pool** , finalists must present a **working prototype** in person.
- **First Prize:** ₹225,000 | **Second Prize:** ₹175,000 | **Third Prize:** ₹100,000.
- **Technical Support** For platform-related issues or submission queries, contact HackerEarth support at **support@hackerearth.com** .